Week 8 — Entanglement and Quantum Communication

EEQT30001 Quantum Technology and Engineering, Unit 2 (Quantum Optics and Quantum Sensing)

Jhih-Sheng Wu 2026-10-27 (3 lecture hours)

Learning objectives (Week 8)

1. Explain how a $\chi^{(2)}$ crystal converts one pump photon into a pair of daughter photons (SPDC), and why the process is parametric amplification of vacuum fluctuations — the same squeeze operator as in Week 7.
2. Compute what a beam splitter does to quantum states in the operator language, derive the Hong–Ou–Mandel dip, and relate its depth to photon indistinguishability.
3. Derive Bell's 1964 inequality and its experiment-ready CHSH form, prove the local-realism bound $|S| \leq 2$, and explain the quantum maximum $2\sqrt{2}$ (Tsirelson).
4. Prove the no-cloning theorem and apply it to quantum communication: BB84 and the 25% intercept–resend error rate, CV-QKD with coherent states, and quantum teleportation.

Reading: Gerry & Knight, *Introductory Quantum Optics*, Ch. 6 (Sec. 1), Ch. 9 (Sec. 2), Ch. 11 (Sec. 3); the 2022 Nobel *scientific background* (public PDF). Full derivations: QO_note_6, Sec. 2.

Opening puzzle. Einstein asked, in effect: does the Moon exist when nobody looks? For thirty years this sounded like philosophy — a matter of taste about what "exists" between measurements. Then, in 1964, John Bell turned it into an inequality [1]: a *measurable* bound that every local, realistic worldview must obey, and quantum mechanics predicts is violated. This week covers the deepest argument in the history of physics, how it was settled in the laboratory (the 2022 Nobel Prize), and how the same entangled photons became a technology: keys that cannot be copied, and states that can be teleported.

1 From the Vacuum to Entangled Pairs: SPDC, Beam Splitters, and Hong–Ou–Mandel

1.1 A $\chi^{(2)}$ crystal turns one photon into two

In a linear medium, modes do not talk to each other. In a non-centrosymmetric crystal (BBO, $LiNbO_3$), the polarization has a quadratic term (quantum_optics_beamer, "Nonlinear Optics"),

$$P_i = \chi^{(1)}_{ij} E_j + \chi^{(2)}_{ijk} E_j E_k. \tag{1.1}$$

The E^2 term couples a mode at ω to a mode at 2ω: two ω photons can merge into one 2ω photon (second-harmonic generation), and — running the film backwards — one pump photon at ω_p can split into two daughter photons, the *signal* and the *idler*. The splitting must respect energy and momentum conservation,

$$\omega_s + \omega_i = \omega_p, \qquad \mathbf{k}_s + \mathbf{k}_i = \mathbf{k}_p, \tag{1.2}$$

so the pairs emerge on cones fixed by phase matching (Fig. 1a). This is *spontaneous parametric down-conversion* (SPDC) — the workhorse photon-pair source of every quantum optics laboratory.

1.2 SPDC is parametric amplification of the vacuum

Quantize the degenerate case ($\omega_s = \omega_i = \omega$, one signal mode a, pump mode a_p). The $\chi^{(2)}$ interaction annihilates one pump photon and creates two signal photons, or vice versa (quantum_optics_beamer, "Generation: Parametric Down-Conversion"):

$$\mathcal{H} = \hbar\omega\, a^\dagger a + \hbar\omega_p\, a_p^\dagger a_p + i\hbar\beta\chi^{(2)}\left(a^2 a_p^\dagger - (a^\dagger)^2 a_p\right), \tag{1.3}$$

with β a mode-overlap parameter. A strong laser pump is a coherent state $|\alpha\rangle$ and is barely depleted, so we may replace the operator a_p by the amplitude α. In the frame rotating at ω (on resonance, $\omega_p = 2\omega$) the evolution operator becomes

$$U(t) = \exp\left[\beta\chi^{(2)} t\left(\alpha^* a^2 - \alpha\,(a^\dagger)^2\right)\right] = S(\xi), \qquad \xi = 2\beta\chi^{(2)}\alpha\, t, \tag{1.4}$$

which is exactly the squeeze operator $S(\xi) = \exp[(\xi^* a^2 - \xi(a^\dagger)^2)/2]$ of Week 7 (QO_note_5, Sec. 2.1): squeezing *requires* the quadratic terms $a^2, (a^\dagger)^2$, i.e. a nonlinear process. What does this amplifier act on when nothing is injected? The vacuum. **An SPDC crystal parametrically amplifies vacuum fluctuations: entangled photon pairs are scooped out of the vacuum. The vacuum is not only non-empty — it is a raw material.** (Vacuum thread, 2/4. In Week 9 squeezed vacuum becomes the input of Gaussian boson sampling; in Week 10 it is injected into LIGO.)

Two consequences of Eq. (1.4) matter for this week. First, in the non-degenerate case (distinct signal and idler modes a_s, a_i) the same algebra gives the *two-mode* squeezed vacuum

$$|\psi\rangle = \frac{1}{\cosh r}\sum_{n=0}^{\infty}(\tanh r)^n\,|n\rangle_s|n\rangle_i, \tag{1.5}$$

whose photon numbers are *perfectly correlated*: detecting an idler photon heralds the presence of exactly one signal photon (at small r) — the heralded single-photon source of Week 7's comparison table. Second, in type-II phase matching the two photons of each pair are orthogonally polarized, one on the H cone and one on the V cone (Fig. 1b). Along the two directions where the cones intersect, neither photon has a definite polarization, yet the polarizations are perfectly anti-correlated — the pair is in the entangled state [3]

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|H\rangle_s|V\rangle_i + e^{i\varphi}|V\rangle_s|H\rangle_i\right). \tag{1.6}$$

Here the subscripts s, i label the two intersection directions, not the cones. This state — produced at megahertz rates on a tabletop — is the entangled pair behind the modern photonic Bell tests of Sec. 2 and the protocols of Sec. 3 (the early tests used atomic-cascade pairs; Delft's loophole-free test used electron spins).

1.3 The beam splitter in operator language

The second tool of this week is almost embarrassingly simple: a half-silvered mirror. Classically, a beam splitter maps two incoming amplitudes onto two outgoing ones. The quantum description replaces amplitudes by mode operators (QO_note_6, Sec. 2): labeling the two inputs a_0, a_1

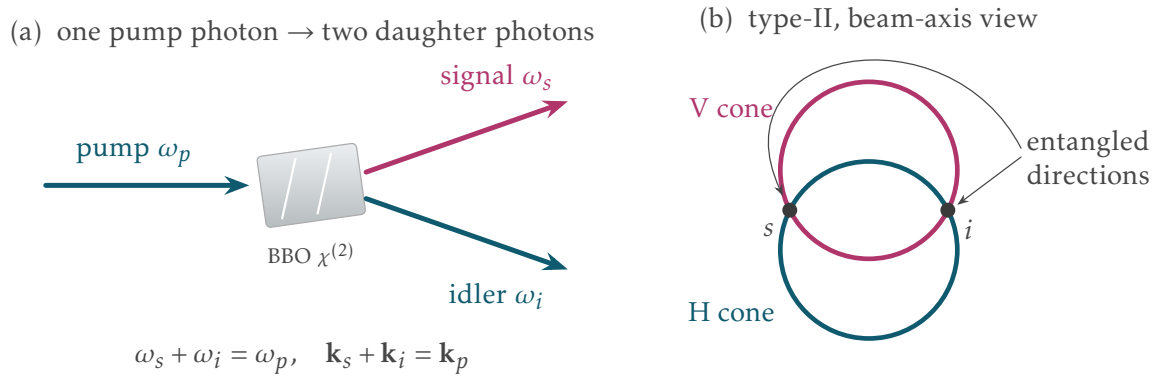


Figure 1: Spontaneous parametric down-conversion. (a) One pump photon splits into a signal–idler pair in a $\chi^{(2)}$ crystal, constrained by energy and momentum conservation, Eq. (1.2). (b) Type-II phase matching: the orthogonally polarized cones intersect along two directions, where the pair is polarization entangled, Eq. (1.6).

and the two outputs a_2, a_3,

$$\begin{pmatrix} a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} t' & r \\ r' & t \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} \equiv U \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}. \tag{1.7}$$

Here r, t are reflection and transmission amplitudes (not to be confused with the squeeze parameter r above). Energy conservation forces U to be unitary: $|r|^2 + |t|^2 = 1$ and $tr^* + t'^* r' = 0$. For a symmetric beam splitter the second condition fixes the relative phase between t and r to $\pm\pi/2$; we adopt the 50:50 convention $t = 1/\sqrt{2}$, $r = i/\sqrt{2}$. Unitarity also guarantees that the outputs remain independent bosonic modes, $[a_i, a_j^\dagger] = \delta_{ij}$: a beam splitter is not a classical coin flip but a *unitary change of mode basis* — nothing is lost, and quantum coherence between the two paths is preserved.

Send in a single photon, $|0\rangle_0|1\rangle_1 = a_1^\dagger|0\rangle$. Transforming the creation operator with U^T gives (QO_note_6, Sec. 2.1)

$$a_1^\dagger|0\rangle \longrightarrow \left(r\,a_2^\dagger + t\,a_3^\dagger\right)|0\rangle = r\,|1\rangle_2|0\rangle_3 + t\,|0\rangle_2|1\rangle_3, \tag{1.8}$$

an entangled state of the two output *modes*: the photon is not "in one arm or the other", it is coherently in both. Closing the two arms with a second beam splitter (a Mach–Zehnder interferometer with path phase θ) converts this coherence into single-photon interference, $P(D_1) = \sin^2\frac{\theta}{2}$, $P(D_2) = \cos^2\frac{\theta}{2}$ — one photon at a time, yet fringes appear. Keep this in mind for Week 10: an interferometer is a machine that converts a phase into a detection probability, and its precision is set by quantum statistics.

1.4 Hong–Ou–Mandel: identical photons always leave together

Now the signature two-photon experiment: one photon into *each* input, $|1\rangle_0|1\rangle_1 = a_0^\dagger a_1^\dagger|0\rangle$. Transforming both creation operators with the 50:50 matrix (QO_note_6, Sec. 2.2),

$$\begin{aligned} a_0^\dagger a_1^\dagger|0\rangle &\longrightarrow \frac{1}{2}\left(a_2^\dagger + i\,a_3^\dagger\right)\left(i\,a_2^\dagger + a_3^\dagger\right)|0\rangle = \frac{i}{2}\left[(a_2^\dagger)^2 + (a_3^\dagger)^2\right]|0\rangle \\ &= \frac{i}{\sqrt{2}}\left(|2\rangle_2|0\rangle_3 + |0\rangle_2|2\rangle_3\right). \end{aligned} \tag{1.9}$$

The $|1\rangle_2|1\rangle_3$ term — classically the most natural outcome — is absent. Its two contributing paths, "both photons transmit" (amplitude $t^2 = \frac{1}{2}$) and "both photons reflect" (amplitude $r^2 = -\frac{1}{2}$), interfere destructively. Two identical photons entering a 50:50 beam splitter from opposite sides *always exit through the same port*: the coincidence rate between the two outputs drops to zero. This is the Hong–Ou–Mandel (HOM) effect [2], and it is a purely quantum interference of *two-photon amplitudes* — there is no classical wave picture for it.

The cancellation in Eq. (1.9) assumed the two photons are *identical*: same frequency, polarization, arrival time, and spatial mode. If their wave packets ϕ_a, ϕ_b are partially distinguishable, only the indistinguishable fraction interferes, and the coincidence probability at zero delay becomes

$$P_{\mathrm{cc}} = \frac{1}{2}\left(1 - |\langle\phi_a|\phi_b\rangle|^2\right). \tag{1.10}$$

Scanning the relative delay τ traces out the celebrated *HOM dip* (Fig. 2b): $P_{\mathrm{cc}} \to \frac{1}{2}$ for fully distinguishable photons (the classical random-routing value), and $P_{\mathrm{cc}} \to 0$ at $\tau = 0$ for identical ones. **The depth of the HOM dip is a direct meter stick for indistinguishability** — which is why it is the standard benchmark for single-photon sources, and why the dip width measures the photon coherence time.

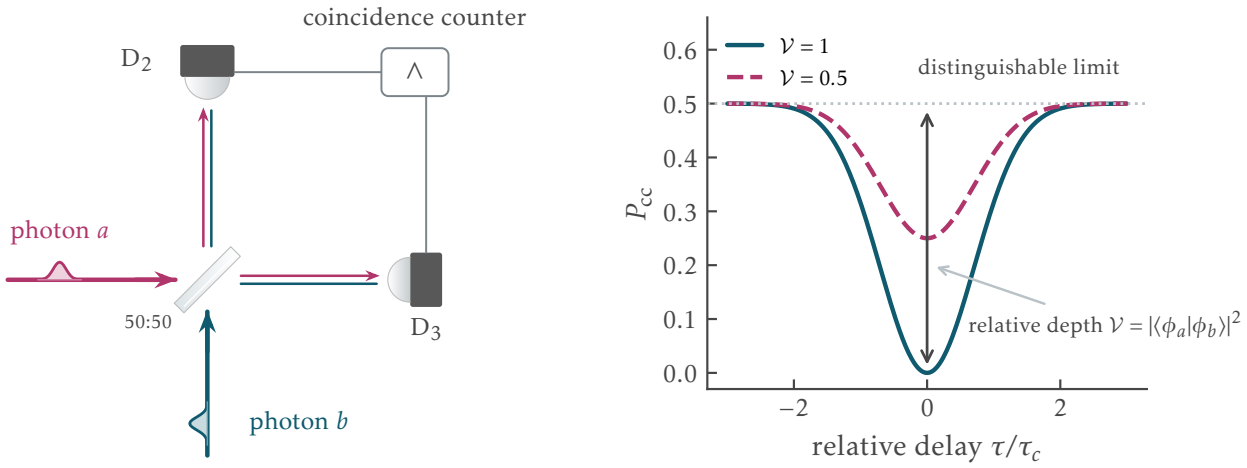


Figure 2: The Hong–Ou–Mandel effect. Left: two single photons meet at a 50:50 beam splitter; a coincidence counter monitors the two outputs. Right: coincidence probability vs. relative delay, Eq. (1.10): identical photons give a dip to zero; partial distinguishability ($\mathcal{V} = |\langle\phi_a|\phi_b\rangle|^2 = 0.5$ shown) makes the dip shallower.

In-class discussion 1. *In a HOM experiment the two photons have slightly different center frequencies. What happens to the dip?* — It becomes shallower (and, for a fixed detection bandwidth, develops a beat): the spectral overlap $|\langle\phi_a|\phi_b\rangle|^2$ decreases, so by Eq. (1.10) the coincidences no longer vanish. Distinguishability in *any* degree of freedom — frequency, polarization, timing — fills the dip back in.

The HOM dip in six lines of QuTiP

```
from qutip import *
from numpy import pi
d = 3
# two modes, up to 2 photons
```

```
a = tensor(destroy(d), qeye(d)); b = tensor(qeye(d), destroy(d))
U = ((1j*pi/4)*(a.dag()*b + a*b.dag())).expm()
# 50:50 beam splitter
out = U * tensor(basis(d,1), basis(d,1))          # |1,1> in
print(expect(a.dag()*a*b.dag()*b, out))
# coincidences -> ~1e-32 (= 0)
```

Summary: Section 1

A $\chi^{(2)}$ crystal couples modes at ω and 2ω; pumped at 2ω it implements the squeeze operator of Week 7 and parametrically amplifies the vacuum, emitting photon pairs that are number-correlated (heralded single photons, Eq. (1.5)) and, in type-II geometry, polarization entangled (Eq. (1.6)). A beam splitter is a unitary change of mode basis, Eq. (1.7); one photon becomes a mode-entangled superposition, and two identical photons bunch, Eq. (1.9) — the HOM dip whose depth measures indistinguishability, Eq. (1.10). SPDC pairs plus beam splitters are the complete toolkit for the Bell tests and protocols that follow.

2 Bell's Theorem: From Philosophy to Measurement

2.1 Einstein's Moon and local realism

Take the entangled pair of Eq. (1.6) with $\varphi = \pi$ — the singlet $|\Psi^-\rangle$. Quantum mechanics says neither photon has a polarization before measurement, yet the two outcomes are perfectly anti-correlated in every basis, no matter how far apart the detectors are. Einstein, Podolsky, and Rosen (1935) found this unacceptable and proposed the "reasonable" alternative: each photon carries hidden instructions fixed at the source (*realism*), and nothing a distant experimenter does can influence the local outcome (*locality*). Correlations would then be no more mysterious than two gloves packed in two boxes. For thirty years the choice between these worldviews was considered metaphysics. Bell's 1964 discovery [1] is that the two worldviews *disagree about measurable numbers* once the analyzers are rotated to intermediate angles.

2.2 From Bell (1964) to CHSH (1969)

Bell's original argument uses the singlet $|\Psi^-\rangle$ ($\varphi = \pi$ in Eq. (1.6)), for which *identical* settings give perfect anti-correlation, $E(a,a) = -1$.[1] In a local-realist model this forces the hidden instructions to be exact anti-copies, $B(x,\lambda) = -A(x,\lambda)$, so that $E(a,b) = -\int d\lambda\, \rho(\lambda) A(a,\lambda) A(b,\lambda)$. Take any three settings a, b, c; using $A^2 = 1$,

$$E(a,b) - E(a,c) = -\int d\lambda\, \rho(\lambda) A(a,\lambda) A(b,\lambda) \Big[1 - A(b,\lambda) A(c,\lambda)\Big]$$

$$\Longrightarrow \quad \big|E(a,b) - E(a,c)\big| \le 1 + E(b,c), \tag{2.1}$$

since the bracket is non-negative and $|A(a)A(b)| = 1$. This is Bell's 1964 inequality [1]. Quantum mechanics violates it: with $E(\theta_a, \theta_b) = -\cos 2(\theta_a - \theta_b)$ and angles $(0^\circ, 30^\circ, 60^\circ)$, the left side equals 1 while the right side is only $\frac{1}{2}$.

[1] For $|\Psi^+\rangle$ ($\varphi = 0$) one finds $E = -\cos 2(\theta_a + \theta_b)$; the violation is the same, only the angle bookkeeping changes.

Beautiful — but not yet an experiment. Equation (2.1) leans on the idealization $E(a,a) = -1$ *exactly*, and no real source or detector delivers perfect anti-correlation. Clauser, Horne, Shimony, and Holt removed that assumption [4], producing the form every experiment actually tests. Alice measures polarization along a or a' and Bob along b or b'; each outcome is ± 1, and $E(a,b)$ is the average of the product — no perfect correlation required anywhere. Define

$$S = E(a,b) - E(a,b') + E(a',b) + E(a',b'). \tag{2.2}$$

In a local-realist world every photon pair carries hidden variables λ that predetermine all four local answers $A(a,\lambda), A(a',\lambda), B(b,\lambda), B(b',\lambda) = \pm 1$. The proof takes two lines: for each λ,

$$A(a,\lambda)\big[B(b,\lambda) - B(b',\lambda)\big] + A(a',\lambda)\big[B(b,\lambda) + B(b',\lambda)\big] = \pm 2, \tag{2.3}$$

because one bracket vanishes and the other is ± 2. Averaging over λ gives $|S| \le 2$ — for *every* conceivable local hidden-variable model, known or not yet invented.

Quantum mechanics disagrees. A half-wave plate at 45° in the idler arm swaps $H \leftrightarrow V$ and a compensator sets $\varphi = 0$, turning the pair of Eq. (1.6) into $|\Phi^+\rangle = (|HH\rangle + |VV\rangle)/\sqrt{2}$; then $E(\theta_a,\theta_b) = \cos 2(\theta_a - \theta_b)$, and the settings $a = 0°$, $b = 22.5°$, $a' = 45°$, $b' = 67.5°$ give

$$S_{\mathrm{QM}} = 2\sqrt{2} \approx 2.83 > 2. \tag{2.4}$$

Figure 3 shows S as the settings are rotated together. The value $2\sqrt{2}$ is itself a ceiling — the *Tsirelson bound* (1980): no quantum state and no measurement can push $|S|$ higher. Nature thus sits in a strange middle band: above everything local realism allows, below what mere non-signaling logic would permit.

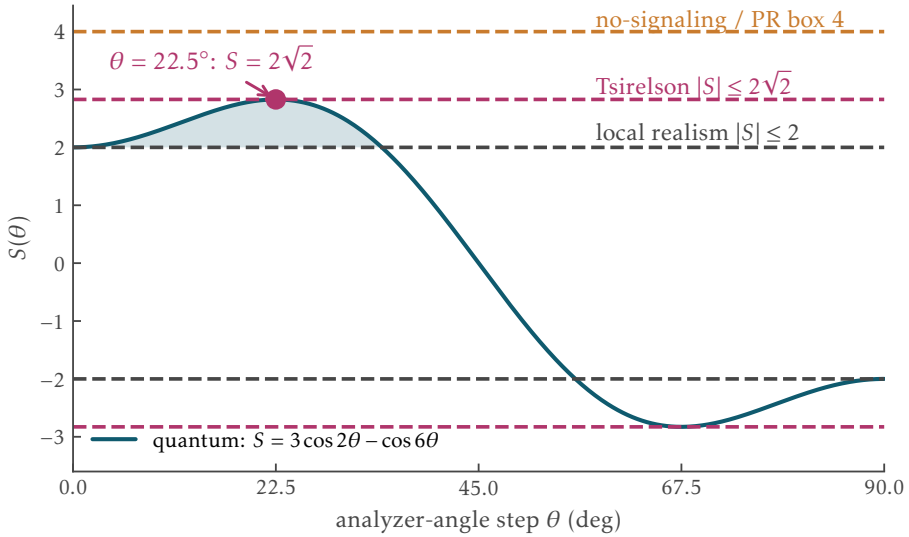


Figure 3: CHSH parameter S for the quantum state $(|HH\rangle+|VV\rangle)/\sqrt{2}$ with settings $(a,a',b,b') = (0, 2\theta, \theta, 3\theta)$, for which $S(\theta) = 3\cos 2\theta - \cos 6\theta$. Local realism caps $|S|$ at 2 (grey); quantum mechanics reaches, but never exceeds, $2\sqrt{2}$ at $\theta = 22.5°$ (Tsirelson bound, magenta).

In-class discussion 2. *An experiment reports* $|S| = 2.9 > 2\sqrt{2}$. *What would that mean?* — Not merely the death of local realism: quantum mechanics *itself* would be refuted, since no quantum state can exceed the Tsirelson bound of Eq. (2.4). (In practice: check the error bars and the analysis before booking a ticket to Stockholm.)

2.3 Sixty years from argument to verdict

Freedman and Clauser made the first measurement (1972) [5]; Aspect and coworkers closed in on locality by switching the analyzers while the photons were in flight [6]; and in 2015 three groups closed the detection and locality loopholes simultaneously — the Delft experiment even used entangled electron spins 1.3 km apart, heralded by photon interference [7]. Every result violated the local bound (Delft: $S = 2.42 \pm 0.20$; NIST and Vienna used the CH–Eberhard form), exactly as quantum mechanics predicts. The 2022 Nobel Prize went to Aspect, Clauser, and Zeilinger for this arc. *In class we watch two short films: the minutephysics–3Blue1Brown video on Bell's theorem (a three-angle, Venn-diagram version of the argument above, with polarizers at* $0°$, $22.5°$, $45°$*) and TU Delft's film on the 2015 loophole-free test. The Nobel scientific background PDF is assigned reading.*

Summary: Section 2

Local realism — outcomes predetermined at the source, no faster-than-light influence — implies Bell's 1964 inequality, Eq. (2.1), under perfect anti-correlation, and with that idealization removed, $|S| \leq 2$ for the CHSH combination, Eq. (2.2) — two lines of arithmetic that never mention quantum mechanics. Entangled photon pairs violate the bound, reaching $S = 2\sqrt{2}$ at the optimal angles (and no more: Tsirelson). Sixty years of experiments, culminating in the loophole-free tests of 2015 and the 2022 Nobel Prize, returned the verdict: the world itself is nonlocally correlated.

3 Quantum Communication: You Cannot Copy a Qubit

3.1 No-cloning in two lines

Suppose a machine could copy an *unknown* quantum state: a unitary U with $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for every $|\psi\rangle$. Apply it to two states $|\psi\rangle, |\phi\rangle$ and take the inner product of the two equations. Unitarity preserves inner products, so

$$\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2 \quad \Longrightarrow \quad \langle\psi|\phi\rangle \in \{0, 1\}\,: \tag{3.1}$$

the machine can at best copy one fixed *orthogonal* set of states (that is just classical copying). No device can clone arbitrary, non-orthogonal quantum states. That is the whole proof — and it has teeth. An eavesdropper cannot copy a qubit in flight and forward the original untouched; and since amplification is cloning, there is no quantum repeater-by-amplifier: extending quantum links beyond fiber attenuation requires entanglement swapping and quantum memories (Week 9).

In-class discussion 3. *Why exactly can an eavesdropper not copy a quantum state and resend it?* — Copying an unknown state outright is forbidden by Eq. (3.1); the only alternative is to *measure* first, but a measurement in the wrong basis disturbs the state, and the disturbance shows up downstream as errors — which is precisely what BB84 exploits.

3.2 BB84: a key whose security is a law of physics

Bennett and Brassard (1984) [8] turned no-cloning into a protocol. Alice encodes random bits in randomly chosen polarization bases: rectilinear $Z = \{|H\rangle, |V\rangle\}$ or diagonal $X = \{|D\rangle, |A\rangle\}$

(45°/135°). Bob measures each photon in his own random basis. Over a public channel they then reveal ***bases only*** and keep the ∼half of the bits where the bases agree (*sifting*); a random subsample of the sifted key is sacrificed to estimate the error rate.

Why is eavesdropping visible? Take the simplest attack: Eve intercepts every photon, measures in a randomly guessed basis, and resends what she found (Fig. 4). With probability $\frac{1}{2}$ she guesses right — the state is undisturbed. With probability $\frac{1}{2}$ she guesses wrong; she resends a photon in the wrong basis, and even when Bob's basis matches Alice's, his outcome is random: error probability $\frac{1}{2}$. The sifted-key error rate is therefore

$$\mathrm{QBER} = \frac{1}{2}\times 0 + \frac{1}{2}\times\frac{1}{2} = 25\%, \tag{3.2}$$

a smoking gun far above any realistic channel noise. Alice and Bob abort; Eve gains a key that is never used. (Practical systems use attenuated laser pulses rather than true single photons — recall Week 7: an attenuated laser is *not* a single-photon source — and defeat photon-number-splitting attacks with decoy states.)

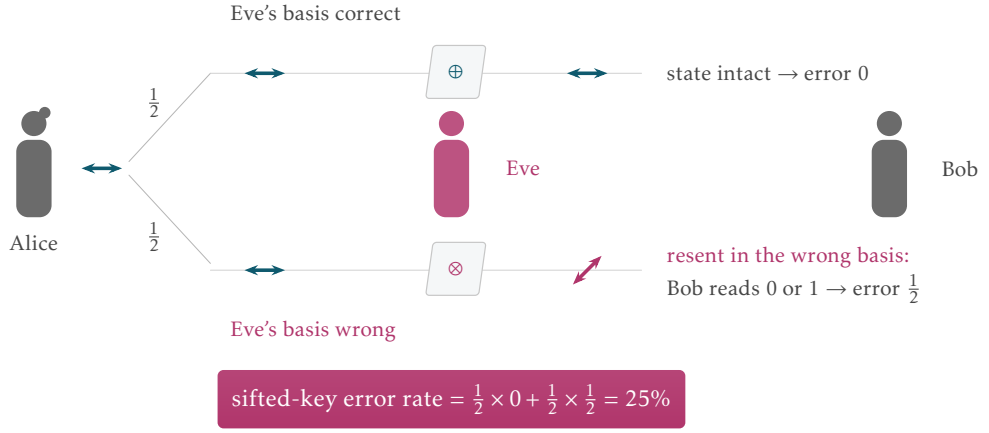


Figure 4: Intercept–resend attack on BB84. Eve's basis guess is right half the time (no disturbance) and wrong half the time (Bob's sifted bit is random), producing a 25% error rate in the sifted key, Eq. (3.2).

3.3 CV-QKD: keys from coherent states

BB84 lives in the DV language. Its CV twin needs no single-photon detector at all: Alice sends *coherent states* displaced by Gaussian-distributed amounts in the quadratures (X, P) of Week 7, and Bob measures a quadrature with a homodyne detector — a beam splitter, a strong local-oscillator laser, and two ordinary photodiodes (Grosshans–Grangier 2002 [9]). Security again rests on the impossibility of copying: non-orthogonal coherent states cannot be cloned, and the uncertainty relation $\Delta X\,\Delta P \geq \frac{1}{2}$ forces any interception to add measurable noise to the quadrature data. The attraction is engineering: CV-QKD runs on standard telecom lasers and detectors at room temperature and coexists with classical traffic in the same fiber; the price is heavier classical post-processing and shorter reach than the best DV links [10]. (Homodyne detection returns, at full strength, in Week 10 — it is how LIGO reads out a squeezed vacuum.)

3.4 Teleportation: moving a state without moving a particle

Can Alice send Bob an *unknown* qubit $|\psi\rangle$ without physically transmitting it — and without violating no-cloning? Yes, if they share one entangled pair in advance [11]. Writing Alice's two particles (the unknown qubit and her half of the pair) in the basis of the four maximally entangled *Bell states*, the joint state rearranges identically into

$$|\psi\rangle_1|\Phi^+\rangle_{23} = \frac{1}{2}\sum_{k=0}^{3}|\mathrm{Bell}_k\rangle_{12}\left(\sigma_k|\psi\rangle_3\right), \tag{3.3}$$

where σ_k are the identity and the three Pauli operators. Alice performs a *Bell-state measurement* (BSM) on particles 1 and 2 and phones Bob the two-bit result k; Bob applies σ_k^{-1} and holds a perfect $|\psi\rangle$. Note the bookkeeping: Alice's measurement destroys her copy (no-cloning is honored), the two classical bits carry no information about $|\psi\rangle$ by themselves, and nothing travels faster than light: by the *no-communication theorem*, no local operation of Alice's can change Bob's reduced density matrix, which stays maximally mixed until the classical message arrives. With photons, the BSM is built from Sec. 1 hardware: the two photons interfere on a 50:50 beam splitter, and HOM logic distinguishes two of the four Bell states — a linear-optics BSM therefore succeeds at most half the time. Bouwmeester *et al.* demonstrated photonic teleportation in 1997 [12]; teleportation of entanglement itself (*entanglement swapping*) is the elementary step of the quantum repeaters and networks of Week 9.

3.5 Coda: what Bell actually constrains

Bell's theorem is often summarized as "quantum mechanics is weird". That undersells it. The inequality $|S| \le 2$ was derived without quantum mechanics; it binds *every* theory in which outcomes are predetermined locally. The experiments say nature violates it. **Bell's theorem constrains not quantum mechanics but all of local realism — whatever physics comes after quantum theory, the world it describes will remain nonlocally correlated.** The same correlations that troubled Einstein are now infrastructure: they certify secret keys, teleport states, and — next week — link quantum processors into networks.

Summary: Section 3

Unitarity forbids copying unknown quantum states, Eq. (3.1). BB84 turns that prohibition into key distribution: intercept–resend leaves a 25% sifted-key error rate, Eq. (3.2). CV-QKD does the same with coherent states, Gaussian modulation, and homodyne readout on telecom hardware. Teleportation transfers an unknown state using one shared entangled pair, a Bell-state measurement (a beam splitter plus HOM interference, success $\le 50\%$ with linear optics), and two classical bits — destroying the original, respecting relativity, and providing the primitive behind entanglement swapping and quantum repeaters.

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